

Elicit vs. Teach, Mechanistically: Eleven Metrics on the TinyStories-1B Twins

MARS V — Mechanistic Understanding of Elicitation vs. Teaching

Working results document, September 2026

Setup

The experiment. Two identical copies (“twins”) of a 1B-parameter language model pretrained only on children’s stories — no arithmetic anywhere in the corpus — are fine-tuned on the same target task: natural-language addition and subtraction of 1–4-digit numbers, **What is the difference between 8 and 5784? $\rightarrow$ -5776.** The training protocol is identical for both. The *only* difference is the checkpoint the fine-tune starts from:

- **Teach twin:** the blank base model. The capability is absent.
- **Elicit twin** (“TS1B-latent”): the blank base plus two installations, neither of which shows the target task: (i) arithmetic in symbols only ($23 + 45 = 68$, 1M examples, full fine-tuning) and (ii) a word $\leftrightarrow$ symbol binding dose that only *rewrites* questions (**What is the sum of 621 and 5068? $\leftrightarrow$ 621 + 5068**) and never shows an answer. This twin can compute in symbols and knows what the words mean, but on the target task it scores 0.000: asked a question, it rewrites it instead of answering. The capability is latent.

Every construction step has a control (Table 1); the two twins are behaviourally identical on the target task before fine-tuning.

Construction step / check	value	control
symbol install: exact match on $23 + 45 =$	0.67–0.73	blank twin: 0.00
corpus census: “sum” in 439M words of stories	$21\times$	“equals” 17, “minus” 57
binding dose: question $\rightarrow$ symbols rewrite exact	0.484	blank + same dose: 0.000
sum-only dose on <i>difference</i> questions	writes “+”	binding is per word
exact match on the target task, before fine-tuning	0.000	= blank twin

Table 1: The elicit twin before fine-tuning: can compute in symbols, understands the words, still answers nothing. Everything below is about what differs inside the two twins and what that difference does.

Endpoints. Seven fine-tuned models enter the results (Table 2). With the paper’s adapter protocol (LoRA), the teach twin stops by the protocol’s own convergence rule far below the elicit twin — 0.093 with 1M examples and 0.139 with four times as many unique examples — which is where the paper’s own Figure 2 puts this model (1–2 nats per token at 1M; “stalls at lower accuracy”). Teaching this base needs full fine-tuning. Two fully fine-tuned teach endpoints are carried through every metric: the *format-installed* twin (the paper’s “pre-teach format” control: the blank base first fine-tuned on a random-label dose in the target’s own format, so it already emits a number after every question; exact match 0.000) fine-tuned to convergence, which reaches 0.424; and the

blank twin fine-tuned to convergence, which reaches 0.819. The elicit twin was fully fine-tuned with the identical recipe (0.986). The format-installed pair is the design-matched comparison — both parents are the blank base plus a full-FT dose, both already output numbers, only one has the algorithm — and the blank-base pair is the better-performing one; the two taught endpoints agree on every metric below. LoRA endpoints are kept as robustness rows; two metrics need adapter snapshots (formation time, weight travel) and exist only for the LoRA runs.

endpoint	parent	method	data	steps (passes)	stop	exact match
elicit LoRA	TS1B-latent	LoRA $r=512$	1M	11,500 (1.5)	converged	0.981
teach LoRA 1M	blank	LoRA $r=512$	1M	25,000 (3.2)	converged	0.093
teach LoRA 4M	blank	LoRA $r=512$	4M	39,500 (1.3)	converged	0.139
teach FT	blank	full FT	4M	62,500 (2.0)	ceiling	0.771
teach FT, converged	blank	full FT	4M	72,000 (2.3)	converged	0.819
teach FT, format-installed	format-installed	full FT	4M	58,500 (1.9)	converged	0.424
elicit FT	TS1B-latent	full FT	4M	21,500 (0.7)	converged	0.986

Table 2: Fine-tuned endpoints. LoRA learning rate 3.53×10^{-4} , full FT 2×10^{-5} ; batch 128; seed 316; training stops when validation loss stops improving (eps/k rule). Exact match = 0-shot accuracy on 1,024 held-out problems. The 4M file is the 1M file plus 3M new unique problems, disjoint from every evaluation set. “Converged” teach FT continues the 2-pass run from its checkpoint until the rule fires (optimizer state restarted).

Two terms used throughout. *Answer position*: the last prompt token, where the model predicts the first token of the answer; every activation measurement is taken there unless stated. *Residual stream*: the model’s running internal state at a token position, a 2048-dimensional vector that every layer reads and adds to. Everything else is defined where it is used.

How to read this document. Eleven metrics, each with the same three blocks: **In one sentence** (what the metric asks), **How it is measured** (the procedure, complete enough to reproduce), **Elicit vs. teach** (the numbers). The metrics are grouped by what they look at: the *cost* of learning (M1), the *machinery* that results (M2–M4), the *dynamics* of building it (M5–M8), and the *latent state* before any target training (M9–M11). Table 11 at the end puts every metric for the full-FT endpoints on one page.

A. Cost of learning

M1 Learning cost — *cheaper with every example vs. the hump*

In one sentence. How many extra nats it costs the model to learn each new example, and whether that cost only falls (connecting something that exists) or first rises (building something new).

How it is measured. Prequential excess description length (EDL), the paper’s learning-cost measure. Train on a stream of n examples; before each example is trained on, record the extra nats the model needs to predict its answer tokens beyond a converged reference; sum over the stream and divide by the number of answer tokens (EDL per token). Repeat on a sweep $n = 100 \dots 4 \times 10^6$ with the identical LoRA protocol (19 rungs per arm, all converged), plus 0-shot exact match on 1,024 held-out problems at each rung. A falling curve means each example costs less than the last; a down–up–down curve (“the hump”) means mid-run examples are the most informative because structure is being built. Three arms: the elicit

twin, the blank teach twin, and the format-installed teach twin. The full-FT endpoints have no sweep; their cost is reported as steps to convergence and held-out test loss.

Elicit vs. teach. Figure 1, Table 3. **Elicit**: strictly monotone decreasing, $4.53 \rightarrow 0.092$ nats; exact match 0.54 already at $n=3,162$, 0.98 at 1M. **Teach** (blank): down-up-down, minimum 0.78 at 14.7K, peak 2.02 at 215K, 1.40 at 1M; exact match 0.093 at 1M and 0.139 at 4M. **Teach** (format-installed): the same hump with the format cost prepaid, minimum 0.64 at 10K, peak 1.33 at 100K, 0.88 at 1M, exact match 0.079. The dataset size at which exact match crosses 0.5 differs by more than $300\times$. Full FT: elicit converges in 21,500 steps (0.7 of a pass, test loss 0.015, exact match 0.986); blank teach runs to the two-pass ceiling (0.771) and converges 9,500 steps later at 0.819 (test loss 0.154); format-installed teach converges after 58,500 steps at 0.424 (test loss 0.639). *Reading*: eliciting spends every example connecting a capability that already exists; teaching has to build it, so the cost rises mid-run and the model converges lower after three times the steps — and the hump survives removing the format, so it is the algorithm being built.

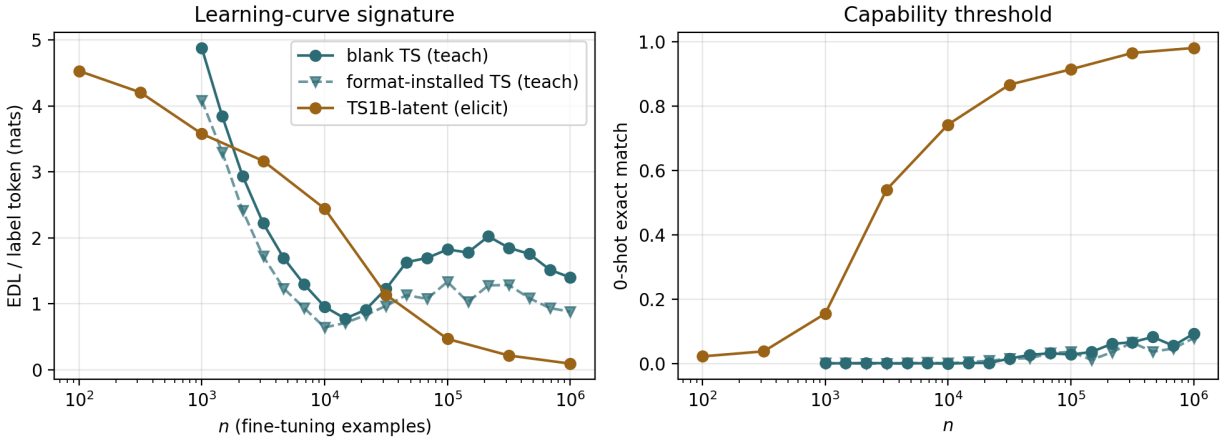


Figure 1: EDL/token (left) and 0-shot exact match (right) vs n ; teal = teach twin from the blank base, dashed teal = teach twin from the format-installed base, gold = elicit twin.

B. The machinery that results

M2 Circuit overlap — *the same circuit vs. a new circuit*

In one sentence. Whether two models solve the task with the same set of components, read against how repeatable a circuit is at all.

How it is measured. *Building a circuit.* The model is split into 528 nodes: 512 attention heads (32 per layer $\times$ 16 layers; heads move information between token positions) and 16 MLP blocks (one per layer; they transform it in place). For each node, attribution patching scores how much the answer’s logit would change if that node’s activation at the answer position were replaced by its value on a different problem: gradient of the answer logit-difference with respect to the node’s activation, times the activation difference between the clean and the corrupted problem, averaged over 128 – 256 problem pairs. The 32 highest-scoring nodes are the circuit. A model that cannot do the task gives a noise map, which the tool flags (its answer logit-difference is ≈ 0). *Comparing circuits.* $\text{Jaccard@32} = \text{shared nodes} \div \text{union of}$

n	elicit (TS1B-latent)		teach (blank)		teach (format-installed)	
	EDL/tok	EM	EDL/tok	EM	EDL/tok	EM
100	4.533	0.023	6.511	0.007	–	–
316	4.208	0.038	6.387	0.033	–	–
1,000	3.579	0.155	4.882	0.001	4.087	0.001
3,162	3.168	0.541	2.221	0.001	1.720	0.000
10,000	2.444	0.743	0.956	0.000	<i>0.641</i> (min)	0.001
14,678	–	–	<i>0.775</i> (min)	0.002	0.708	0.005
31,623	1.137	0.867	1.228	0.016	0.968	0.016
100,000	0.467	0.915	1.824	0.028	<i>1.333</i> (peak)	0.037
215,443	–	–	<i>2.022</i> (peak)	0.062	1.279	0.035
316,228	0.215	0.966	1.849	0.066	1.285	0.065
1,000,000	0.092	0.981	1.399	0.093	0.880	0.079
4,000,000	–	–	0.930	0.139	–	–

Table 3: Per-rung values, LoRA protocol (EDL in nats per answer token; EM = exact match; 19 rungs per teach arm, all converged; rungs shared with the elicit sweep shown plus each arm’s minimum and peak).

the two 32-node sets. Two random circuits overlap at 0.031. The *ceiling* is the overlap of two circuits of the *same* model computed from two disjoint halves of the data — how repeatable a circuit is (0.42–0.68 here); an overlap at the ceiling means “the same circuit”. Spearman = rank correlation of the node scores over the union. Also the layer histogram of each circuit’s top-16 nodes, and for the LoRA-1M pair the change at the level of connections between nodes (edge-level $\Delta S = 1 - \text{Jaccard}$ over the 256 strongest node→block edges).

Elicit vs. teach. Table 4, Figure 2. **Parent** → **child**. The elicit parent’s symbol circuit ↔ its child’s natural-language circuit: 0.455–0.524 (about 70% of the ceiling), and the top four nodes are the same four in the same order (`mlp:15,14,13,12`); the shared backbone is complete at $n=3K$ (`child@3K` ↔ `child@1M`: 0.391). The blank parent has no measurable circuit (logit-diff -0.33 even with 16 examples in the prompt), so the taught child has nothing to reuse. **Child** ↔ **child**. The two elicited children — one LoRA, one full FT — overlap at 0.641, at the ceiling: one circuit whichever way it is trained. Every taught child sits apart from them: 0.306 for the format-installed full-FT pair (Spearman 0.03), 0.333 for the blank-base full-FT pair, 0.391 for the LoRA-4M pair, 0.231 for the LoRA-1M pair — about 60% of the ceiling, score correlations between -0.11 and 0.28 . Taught children agree with each other at 0.36–0.60. What all circuits share is the late-layer MLP engine (`mlp:15,14,13,7`); what differs is the rest: taught circuits add `mlp:0,10,11` (the format-installed one also `mlp:1,3,4,5,6`) and their own attention heads (eight layer-0 heads over the raw digits under LoRA; heads in layers 4–7 and 15 under full FT), elicited circuits use `mlp:8,9` and three layer-7 heads. LoRA-1M pair: $\Delta S_{\text{node}} = 0.769$, $\Delta S_{\text{edge}} = 0.739$, both far above the noise floors (0.32 / 0.61). *Reading*: elicitation re-weights the circuit the parent already had; teaching builds a related but different one — the same late-layer MLPs, its own way of fetching and routing the operands.

M3 Circuit faithfulness — *does the circuit carry the computation?*

In one sentence. Whether the 32 nodes called “the circuit” really produce the answer — the check that licenses every overlap in M2 as a comparison between two real mechanisms.

How it is measured. True activation patching, in two directions. *Sufficiency*: run a cor-

Comparison	Jaccard@32	chance	ceiling(s)
parent’s symbol circuit $\leftrightarrow$ elicited child (two eval surfaces)	0.455–0.488 / 0.524	0.031	0.684
elicited child @3K $\leftrightarrow$ @1M	0.391	0.031	0.684
elicited child, full FT $\leftrightarrow$ elicited child, LoRA	0.641 (Spearman 0.54)	0.031	0.561 / 0.684
blank parent $\leftrightarrow$ taught child	undefined (parent map is noise)	–	–
elicited FT $\leftrightarrow$ taught FT, format-installed	0.306 (Spearman 0.03)	0.031	0.561 / 0.524
elicited FT $\leftrightarrow$ taught FT, blank, 2 passes / converged	0.333 (Spearman 0.16) / 0.333 (0.28)	0.031	0.561 / 0.422
taught FT, converged $\leftrightarrow$ taught FT before continuation	0.600 (Spearman 0.57)	0.031	0.422 / 0.422
taught FT, format-installed $\leftrightarrow$ taught FT, blank, converged	0.362 (Spearman 0.18)	0.031	0.524 / 0.422
elicited $\leftrightarrow$ taught, LoRA pair (4M)	0.391 (Spearman 0.28)	0.031	0.684 / 0.524
elicited $\leftrightarrow$ taught, LoRA pair (1M)	0.231 (Spearman -0.11)	0.031	0.684 / 0.561
taught, full FT $\leftrightarrow$ taught, LoRA 4M	0.391 (Spearman 0.11)	0.031	0.422 / 0.524

Table 4: Circuit overlaps. All maps pass the performing-regime check.

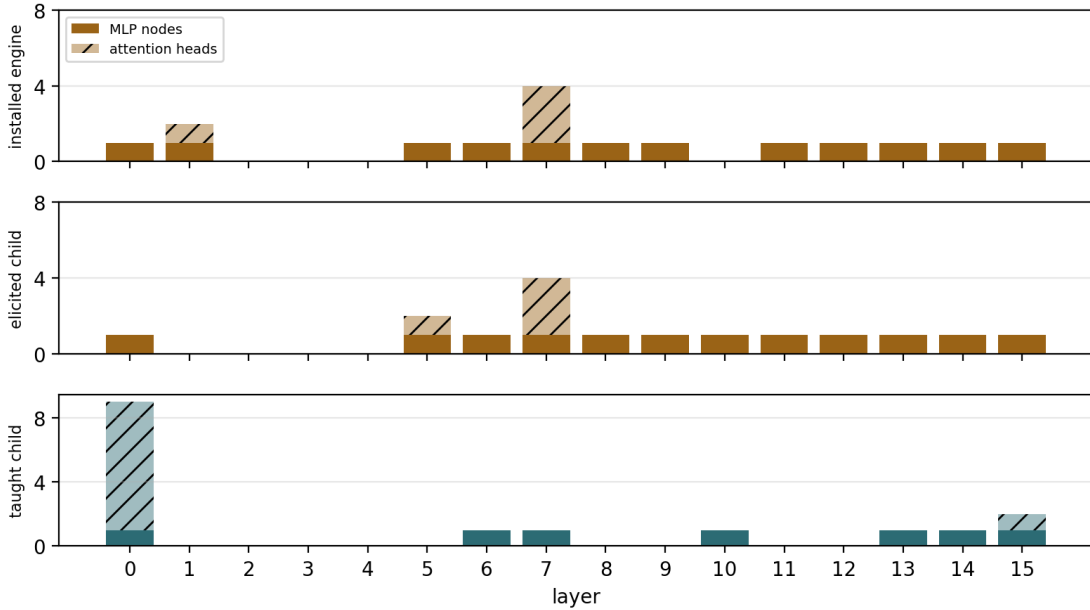


Figure 2: Top-16 node layer histograms (solid = MLP, hatched = attention) for the parent’s symbol circuit, the LoRA-elicited child and the LoRA-1M taught child.

rupted problem, copy the clean problem’s activations into only the top- k nodes, and measure what fraction of the clean-vs-corrupt answer logit-difference is recovered. *Necessity*: run the clean problem, corrupt only the top- k nodes, and measure what fraction is destroyed. 128 problem pairs, $k = 8, 16, 32, 64, 128, 528$; 1.0 means the top- k nodes alone account for the whole behaviour.

Elicit vs. teach. Top-8 of 528 nodes recover / destroy the answer at: LoRA-elicited 0.997 / 0.998; full-FT elicited 0.923 / 0.943; full-FT taught 0.950 / 0.966 (converged 0.950 / 0.965; format-installed 0.983 / 0.988); LoRA-4M taught 0.976 / 0.987. Top-32 ≥ 0.996 everywhere. *Reading*: the regimes do not differ here — both build compact circuits that are near-totally sufficient and necessary — so the differences in M2, M4 and M5 are differences between two load-bearing circuits, not between a circuit and noise.

M4 Head roles — *the same reader heads vs. rediscovered ones*

In one sentence. Which attention heads carry each piece of the problem — the first number, the second number, the operation — and whether a fine-tuned model keeps the parent’s readers or has to find its own.

How it is measured. Desiderata Component Masking (Prakash et al. 2024). For one piece of information at a time, take a problem and a counterfactual problem that differs only in that piece, and learn (by gradient descent on a mask over heads, with a sparsity penalty) the smallest set of attention heads such that copying those heads’ activations from the counterfactual run into the clean run makes the model answer the counterfactual problem. That set is the heads carrying that piece (its “role”). The set is validated by its counterfactual-flip accuracy against the ceiling obtained by copying every head; a set is “compact” only if it reaches that ceiling. Role sets are compared across models by Jaccard (chance ≈ 0.05); 64–128 counterfactual pairs, heads only.

Elicit vs. teach. Table 5. **Elicit:** parent $\leftrightarrow$ LoRA-elicited child, flip accuracy at the ceiling for every set, first operand 0.879 (29 of 33 heads shared), second operand 0.900, operation 0.750; the operand heads are almost all layer-0 attention (30 of 32). The full-FT elicited child keeps the same heads (0.811 / 0.833 vs the LoRA child, 0.694 / 0.857 vs the parent). **Teach:** every performing taught model has compact operand sets at its ceiling, overlapping the parent’s engine at 0.76–0.88 (first operand) / 0.66–0.68 (second) and the elicited child’s at 0.59–0.66 / 0.69–0.76; the full-FT taught set contains 28 of the full-FT elicited child’s 31 first-operand heads and adds nine more. The two LoRA-taught children that never learned the task (exact match 0.09 / 0.14) have no compact set at all: best flip accuracy 0.008–0.04 against ceilings of 0.14–0.20. **Reading:** operand reading is a layer-0 function of the substrate, so every model that solves the task uses largely the same reader heads; elicitation keeps the parent’s set nearly intact (0.88–0.90), teaching rediscovered it and lands on a larger set (0.6–0.88) — a graded difference.

Comparison (first operand / second operand)	$ A $	$ B $	Jaccard	flip acc. / ceiling
parent $\leftrightarrow$ elicited (LoRA), symbol surface	30 / 27	32 / 30	0.879 / 0.900	at ceiling
parent $\leftrightarrow$ elicited (LoRA), operation	25	24	0.750	at ceiling
elicited (full FT) $\leftrightarrow$ elicited (LoRA)	31 / 25	36 / 30	0.811 / 0.833	0.98 / 0.98
elicited (full FT) $\leftrightarrow$ parent	31 / 25	30 / 27	0.694 / 0.857	–
taught (full FT) $\leftrightarrow$ elicited (LoRA)	37 / 37	36 / 30	0.587 / 0.763	0.81 / 0.79 (0.81 / 0.80)
taught (full FT) $\leftrightarrow$ elicited (full FT)	37 / 37	31 / 25	0.700 / 0.632	–
taught (full FT) $\leftrightarrow$ parent	37 / 37	30 / 27	0.763 / 0.684	–
taught (FT, format-installed) $\leftrightarrow$ parent	32 / 41	30 / 27	0.879 / 0.659	0.59 / 0.54 (0.59 / 0.55)
taught (FT, format-installed) $\leftrightarrow$ elicited (LoRA)	32 / 41	36 / 30	0.659 / 0.690	–
taught (FT, blank, converged) $\leftrightarrow$ parent	37 / 36	30 / 27	0.763 / 0.658	0.85 / 0.81 (0.85 / 0.83)
taught (FT, blank, converged) $\leftrightarrow$ elicited (LoRA)	37 / 36	36 / 30	0.587 / 0.737	–
taught (LoRA 4M) $\leftrightarrow$ elicited (LoRA)	12 / 21	36 / 30	0.043 / 0.041	0.01 / 0.03 (0.20 / 0.14)
taught (LoRA 1M) $\leftrightarrow$ elicited (LoRA)	9 / 21	36 / 30	0.125 / 0.085	0.02 / 0.04 (0.15)

Table 5: Head roles by Desiderata Component Masking. $|A|$, $|B|$ = size of the two role sets; flip accuracy is reported for set A .

C. Dynamics: how the machinery comes into being

M5 Circuit formation time — *present at step 1 vs. assembled mid-run*

In one sentence. When during training the final circuit comes into existence.

How it is measured. Adapter snapshots were saved along the LoRA runs (128 per run, log-spaced). For each snapshot, build the circuit exactly as in M2 and take its Jaccard@32 with the run’s final circuit; read against chance (0.031) and the split-half ceiling. The step at which the curve leaves the noise floor is when the circuit starts to exist; the step at which it reaches the ceiling is when it stops changing.

Elicit vs. teach. Figure 3. **Elicit:** 0.488 at step 1 ($16\times$ chance), a brief dip to ~ 0.42 while the input interface is wired (step 154), then locked from about step 1.5K of 11.5K (0.561; 82% of the ceiling, 97% of the Jaccard@64 ceiling by 8.4K). **Teach** (LoRA 1M): at the noise floor (0.22–0.28) until about step 1.3K, then $0.27 \rightarrow 0.44$ over steps 1.3K–5.4K — the region where its EDL peaks (M1). Teach (LoRA 4M): about a third of the endpoint set from step 1 (0.33–0.46 against a 0.52 ceiling), the rest churning to the end (0.36 at 8K and 37K; logit-diff $0.9 \rightarrow 13.6$). *Reading:* the elicited circuit is present before training starts and stops changing by one-eighth of the run; the taught circuit does not exist until a thousand steps in and is built exactly while the learning cost is rising.

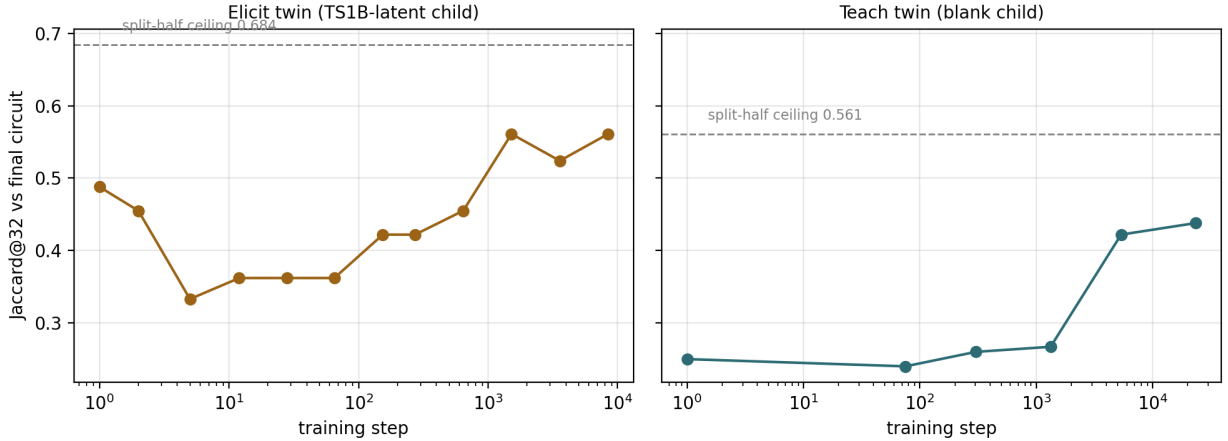


Figure 3: Circuit formation, elicit (left) vs teach (right); dashed = split-half ceiling.

M6 Gradient pressure — a dying gradient vs. a growing one

In one sentence. How hard training is pushing on the weights, and whether that push fades (nothing left to change) or grows (structure still being built).

How it is measured. The global gradient norm before clipping, logged at every update. Reported: its peak and the step of the peak; its mean over the first 1%, the middle, and the last 10% of steps; the ratio first-1% $\div$ last-10% (above 1 = decays, below 1 = grows); and the cumulative gradient mass $\sum_t \|g_t\|$. For the LoRA runs also how that mass splits across attention query/key weights (QK), attention value/output weights (VO) and MLP weights; the full-FT trainer logs the global norm only. Norms are comparable within a method, not across methods (different parameter sets).

Elicit vs. teach. Table 6, Figure 4. **LoRA.** Both arms start at the same scale (first-1% mean 0.18 vs 0.17). Elicit decays $3.3\times$ to a floor of 0.056 (peak 0.44 at step 96). Teach grows $40\text{--}50\times$: last-10% mean 6.8 at 1M and 7.8 at 4M, peaks of 17.4 and 22.0 near the end.

Cumulative mass 783 vs 116,877 (1M) and 220,685 (4M): 150–280 $\times$. Across the sweep the arms are indistinguishable up to $n=10\text{K}$ and diverge from 31.6K. Teach’s mass moves out of the MLPs (share 0.47 $\rightarrow$ 0.01) into attention (VO share 0.87 at 1M, 0.93 at 4M). **Full FT.** Elicit peaks at step 1 (34.0) and decays 3.4 $\times$ to a floor of 1.4. Teach from the blank base starts lower (2.3), peaks at step 3,844 (32.0), holds 5.3 mid-run and ends at 3.4 (1.5 $\times$ its start); mass 353,277 vs 44,021 (8 $\times$; 2.8 $\times$ per step over 2.9 $\times$ the steps); the continuation adds 9,500 steps at a flat norm (30,954 more). Teach from the format-installed parent: 2.69 at the start, peak 70.9 at step 50,860 (the largest single update anywhere), 3.77 at the end (1.4 $\times$); mass 378,170 (8.6 $\times$ elicit). *Reading:* elicitation converges and its gradient dies, by the same 3.3–3.4 $\times$ under either method; teaching generates gradient pressure that grows with the structure being built — 40–50 $\times$ under LoRA, 1.4–1.5 $\times$ under full FT — and under LoRA aims it at the attention weights where the operand-fetching heads of M4 are being written.

	elicit LoRA 1M	teach LoRA 1M	teach LoRA 4M	elicit FT	teach FT, blank (+ cont.)	teach FT, format-inst.
steps	11,500	25,000	39,500	21,500	62,500 (+ 9,500)	58,500
peak gradient norm (step)	0.441 (96)	17.4 (23,657)	22.0 (36,931)	34.0 (1)	32.0 (3,844)	70.9 (50,860)
mean norm: first 1% / middle / last 10%	0.183 / 0.060 / 0.056	0.169 / 5.21 / 6.80	0.160 / 6.61 / 7.78	4.80 / 1.85 / 1.39	2.30 / 5.31 / 3.43	2.69 / 4.86 / 3.77
first 1% $\div$ last 10%	$\times 3.3$ (decays)	$\times 0.02$ (grows $\sim 40\times$)	$\times 0.02$ (grows $\sim 50\times$)	$\times 3.4$ (decays)	$\times 0.7$ (grows 1.5 $\times$)	$\times 0.7$ (grows 1.4 $\times$)
cumulative gradient mass $\sum_t \ g_t\ $	783	116,877	220,685	44,021	353,277 (+ 30,954)	378,170
gradient mass per step	0.068	4.68	5.59	2.05	5.65 (3.26)	6.46
mass share QK / VO / MLP	.18 / .55 / .27	.12 / .87 / .01	.05 / .93 / .01	–	–	–

Table 6: Raw gradient strength (pre-clip global norm at every update). LoRA and full-FT norms are on different parameter sets and are compared within a method only.

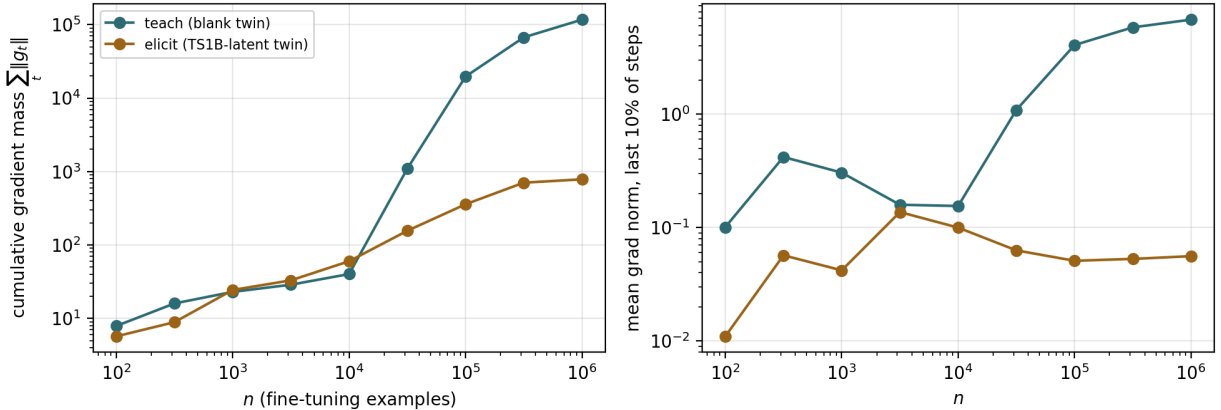


Figure 4: Cumulative gradient mass (left) and late-run gradient norm (right) vs n ; teal = teach, gold = elicit.

M7 Weight write — a small update vs. a large one

In one sentence. How much fine-tuning changes the weights, how far they travel to get there, and how much capability each unit of change buys.

How it is measured. Relative weight change $\|\Delta W\|/\|W\|$ between child and parent — exact from the LoRA factors ($\Delta W = BA$), or from the checkpoint difference for full fine-tuning — over all attention and MLP matrices, with the effective rank of ΔW (participation ratio of its spectrum; 2048 = full rank). Along the LoRA runs, from the adapter snapshots: total travel $\|\Delta W_t\|_F$ vs step, its peak and final speed per step, and capability per unit travel = final exact match $\div$ total travel.

Elicit vs. teach. Table 7. **LoRA**: relative write 0.212 (teach 1M) and 0.264 (teach 4M) vs 0.095 (elicit), 2.2–2.8 $\times$; total travel 457 vs 120 (3.8 $\times$) over twice as many steps; capability per unit travel 2.0×10^{-4} vs 8.1×10^{-3} ($\sim 40\times$). **Full FT**, no adapters involved: 0.094 (teach, blank, 2 passes), 0.102 (converged), 0.086 (format-installed parent) vs 0.050 (elicit), 1.7–2.0 $\times$, with the same effective rank of the change (480–580 vs 554). The travel *profile* is the same in both arms (a burst, then decay): the optimiser’s normalisation compresses the $150\times$ gradient gap of M6 into a 3.8 $\times$ step-size gap. *Reading*: teaching writes two to four times more into the weights than elicitation and gets forty times less capability per unit written, under both methods — eliciting connects what exists, teaching writes what does not.

	elicit (TS1B-latent)	teach (blank)
relative write $\ \Delta W\ /\ W\ $, LoRA 1M	0.095	0.212
relative write, LoRA 4M	–	0.264
relative write, full FT (4M): blank 2 passes / converged / format-installed	0.050	0.094 / 0.102 / 0.086
effective rank of ΔW , full FT (of 2048)	554	571 / 580 / 481
steps to convergence (LoRA 1M)	11,500	23,496
peak / final writing speed per step (LoRA 1M)	0.218 / 0.018	0.306 / 0.022
total weight travel $\ \Delta W\ _F$ (LoRA 1M)	120	457
final exact match (LoRA 1M)	0.981	0.093
capability per unit travel (LoRA 1M)	8.1×10^{-3}	2.0×10^{-4}

Table 7: Weight-space write, elicit vs teach (travel curves in 05_weights/fig_weight_travel).

M8 State-shift direction — a different vector per problem vs. one shared vector

In one sentence. Whether fine-tuning changes the model’s internal state by something problem-specific (the answer, routed to the output) or by the same push on every problem (moving the model into a new mode).

How it is measured. Run the same 256 problems through parent and child (both in fp32) and take the difference of their residual streams at the answer position, layer by layer. *Size*: its norm relative to the parent’s state. *Direction*: stack the 256 difference vectors and compute the fraction of their energy in the single best direction (PC1; 1.0 means every problem’s difference is the same vector). The reference for “unrelated vectors” is the PC1 of the parent’s own 256 states, given alongside. The same PC1 is recomputed after projecting out the directions that encode the answer token and the parent’s top-1 token, so that “the same vector” cannot simply be the answer. Also the cosine of each shift to the mean shift, and, as a functional counterpart, $\text{KL}(\text{parent}||\text{child})$ at the answer position vs on generic story text.

Elicit vs. teach. Figure 5, Table 8. **Elicit**: PC1 of the shift at the read-out 0.063 (LoRA) and 0.056 (full FT), with the parent’s own states at 0.23 — no shared direction. **Teach**: 0.528 (LoRA 4M), 0.521 / 0.526 (full FT from the blank base, 2 passes / converged) and 0.567 from the format-installed parent, still 0.52–0.56 after the answer-token directions are removed. The format-installed parent already emits a number after every question, so its child has no output format to learn — and its shift carries the largest shared component: the component is the algorithm’s mode, not the format. Cosine to the mean shift: elicit 0.26 / 0.23, teach 0.26 / 0.74 / 0.80. Size and depth follow the fine-tuning method, not the regime (LoRA-taught final-layer write 14–29 $\times$ the parent’s norm, full-FT 1.2–1.4 $\times$, elicit 1.7–1.9 $\times$). *Reading*: elicitation changes the state by a problem-specific vector because the parent already sits in a problem-specific state; teaching’s change is dominated by one shared component because the blank

parent’s state at the answer position carries no problem information and the model has to be moved into answering mode before any content can be added.

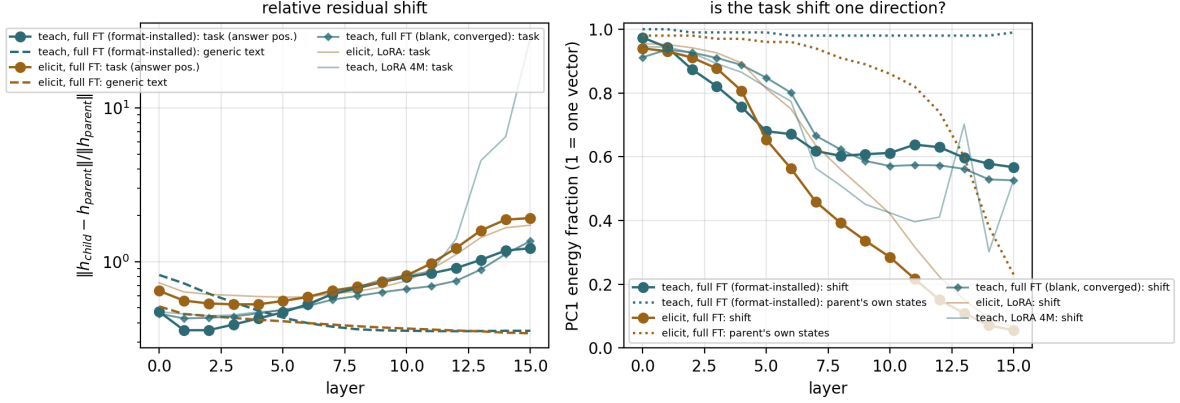


Figure 5: Full-FT endpoints (thick: elicit vs format-installed teach; diamonds: blank-base teach converged) with the LoRA endpoints thin. Left: relative residual shift by layer at the answer position (solid) and on generic story text (dashed). Right: PC1 energy fraction of the 256 per-problem shift vectors, with the parent’s own states dotted.

	elicit LoRA 1M	teach LoRA 4M	elicit FT	teach FT, blank (conv.)	teach FT, format-inst.
final exact match	0.981	0.139	0.986	0.819	0.424
PC1 of shift at the read-out (parent’s own states)	0.063 (0.23)	0.528 (0.98)	0.056 (0.23)	0.526 (0.98)	0.567 (0.99)
PC1 after removing answer-token directions	0.072	0.520	0.061	0.528	0.561
cosine of each shift to the mean shift	0.26	0.26	0.23	0.74	0.80
shift at answer position, final layer (relative)	1.72	28.6	1.92	1.36	1.22
KL(parent child) at the answer position (nats)	14.3	14.0	16.4	22.0	12.3
KL(parent child) on stories (nats/token)	0.067	0.326	0.721	1.790	1.568

Table 8: Residual-stream shift parent→child at the answer position, 256 problems, fp32. Blank-base teach FT before continuation: PC1 0.521, final-layer shift 1.42; LoRA-1M teach: PC1 0.351, shift 14.1.

D. The latent state: what it means and what it permits before training

M9 Latent reach — *do the words already reach the machinery?*

In one sentence. Whether a natural-language question, before any target training, puts the model’s arithmetic components into the same problem-specific state that the symbols do — and whether the model’s state is problem-specific at all.

How it is measured. Render each problem two ways — What is the sum of 621 and 5068? and $621 + 5068 =$ — and record every node’s activation at the answer position. Per node: the mean cosine between the two renderings of the *same* problem minus the mean cosine between renderings of *different* problems (the “bridging index”). Positive means the words put the node into the state the symbols put it in. MLP blocks only: attention heads share digit tokens across renderings and would score trivially. Compared between the symbol-only engine (no binding dose) and the elicit parent (binding dose installed). A second, geometry-only

check: how problem-specific the answer-position state itself is, as PC1 and cosine-to-mean across the 256 problems (≈ 1 means the same state for every problem).

Elicit vs. teach. Bridging index. Symbol-only engine: ≈ 0.00 – 0.01 at every layer. Elicit parent: $+0.07$ – 0.13 across layers 7–15, about $10\times$. **State geometry at the final layer.** Elicit parent: cosine-to-mean 0.51, PC1 0.23 (problem-specific). Blank base: 0.99 and 0.98; format-installed base: 0.99 and 0.99 — the same state for every problem (emitting a number is not knowing which). *Reading:* before any target training, a natural-language question already activates the elicit parent’s arithmetic MLPs and leaves it in a problem-specific state; the teach parent has no arithmetic MLPs to activate and is in the same state for every problem.

M10 Answer depth — *the answer already there vs. absent; finished one layer earlier vs. later*

In one sentence. At which layer the answer becomes readable from the model’s internal state — before training (is it there at all?) and after (how early is it finished?).

How it is measured. Read the model’s intermediate state at each layer as if it were the output, three ways. *Logit lens:* apply the final unembedding to the layer- ℓ state. *J-lens* (Jacobian lens): first transport the state to the final-layer basis with the average Jacobian $\partial h_{\text{final}}/\partial h_\ell$ over held-out story text, then unembed — this reads what the state is *disposed to make the model say*, and is where the answer can appear before the model emits it. *R-lens:* the same transport with a relevance-propagation backward pass, more faithful at early layers. Scored token = the first digit chunk of the answer (for negative answers the sign is fed in first, so predicting a minus sign never counts). Per layer: the answer token’s median rank among 128K vocabulary entries (chance 64K); the *logit-diff*, the model’s preference for the correct answer over a mismatched problem’s answer, in nats (0 = no answer-specific signal); and the *settled depth*, the first layer from which the token stays top-ranked to the end. Rank without logit-diff can be a number prior (any number ranks well in a model that emits numbers); the two are read together.

Elicit vs. teach. Figure 6, Table 9. **Before training.** Elicit parent on the NL task, J-space rank of the answer 2,606 at layer 12, 510 at layer 14, 174 in the actual output, preferring the correct answer by 2.1 nats — while it emits a rewrite, not a number. Blank base: 55–80K at every layer in all three lenses, logit-diff 0.0. Format-installed base: rank 452 at the output because it emits numbers, logit-diff 0.0 at every layer — “a number is coming” without “this number”. **After training.** The LoRA-elicited child ranks the answer 139 / 6 / 1 at layers 12 / 13 / 14 and settles at layer 14 — the parent’s own trajectory on symbols (122 / 6 / 1). Full-FT elicited: 338 / 8 / 1, settled at layer 14 (quartiles 14–14), top-1 0.80 at layer 14. Full-FT taught, blank base: 4,196 / 1,916 / 79, settled at layer 15 (quartiles 15–15), top-1 0.12 at layer 14 rising to 0.80 at 15 (converged: 4,123 / 1,832 / 138, $0.16 \rightarrow 0.84$); from the format-installed parent: 27,044 / 15,058 / 961, settled at layer 15, $0.11 \rightarrow 0.54$. LoRA-taught children: 17–20K at layer 14, then 6–8 at layer 15. *Reading:* the elicit parent carries the answer in its verbalizable space before any target training and elicitation only amplifies that read-out; the blank base carries nothing, and a taught model computes the answer one layer later than the engine even when it solves the task.

M11 Switch-on — *can the capability be turned on without training?*

In one sentence. Whether the task can be unlocked in the parent with its weights frozen — possible only if the machinery is already there.

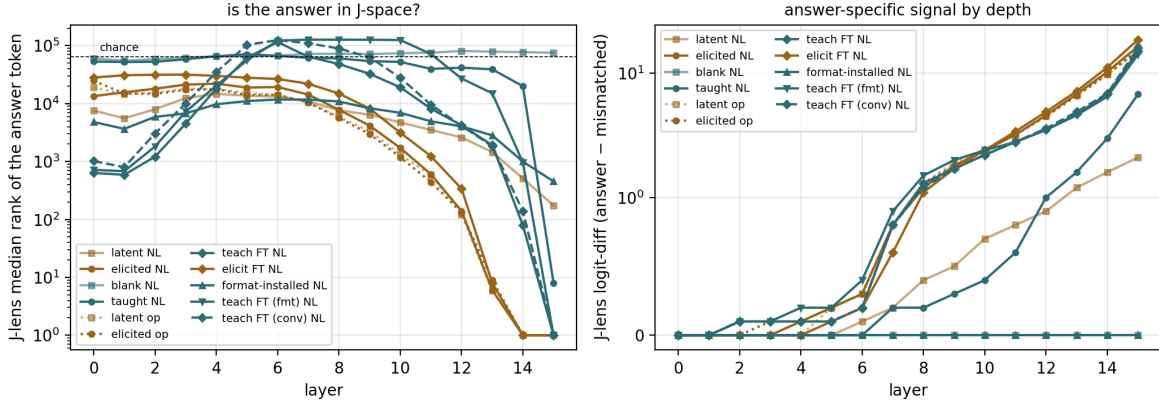


Figure 6: J-lens read-out of the answer token by layer: median rank (left, chance 64K) and logit-diff against a mismatched problem’s answer (right). Squares = parents, triangles = the format-installed base and its child, circles = LoRA children, diamonds = full-FT children; dotted = symbol surface.

model	surface	own acc	rank L12 (logit / J / R)	rank L14 (logit / J / R)	read-out	ld
elicit parent (TS1B-latent)	NL	0.047	5,704 / 2,606 / 2,730	436 / 510 / 424	174	+2.1
blank base	NL	0.000	76,049 / 80,840 / 78,243	77,707 / 77,320 / 79,116	75,441	0.0
format-installed base	NL	0.004	8,394 / 4,036 / 5,417	1,106 / 997 / 998	452	0.0
elicted child (LoRA)	NL	0.957	1,584 / 139 / 464	1 / 1 / 1	1	+16.5
elicted child (full FT)	NL	0.988	1,797 / 338 / 492	1 / 1 / 1	1	+18.7
taught child (full FT, blank, converged)	NL	0.844	18,988 / 4,123 / 9,620	170 / 138 / 459	1	+16.1
taught child (full FT, format-installed)	NL	0.535	23,909 / 27,044 / 25,529	381 / 961 / 1,848	1	+14.2
taught child (LoRA 4M)	NL	0.117	54,896 / 41,722 / 41,666	40,496 / 17,183 / 21,998	6	+7.6
taught child (LoRA 1M)	NL	0.109	51,167 / 41,677 / 44,335	35,620 / 19,933 / 21,156	8	+6.8
elicit parent (TS1B-latent)	symbols	0.836	1,224 / 122 / 334	1 / 1 / 1	1	+15.1
elicted child (LoRA)	symbols	0.855	1,360 / 131 / 399	1 / 1 / 1	1	+15.5

Table 9: Median rank of the answer’s first digit token among 128K entries under the three lenses; “own acc” = the model’s own first-digit-token accuracy on the 256 problems; read-out = the last layer, where all lenses equal the model’s output; ld = logit-diff at the read-out (nats). Blank-base teach FT before continuation: 18,502 / 4,196 / 9,429, then 141 / 79 / 377, own acc 0.797, ld +15.2.

How it is measured. Exact match on the target task under interventions that train nothing. (a) *Self-chain*: let the parent rewrite the question in its own symbols, then feed it its own rewrite. (b) *Donor patching*: take the fine-tuned child’s activations at the 32 circuit nodes (M2) and write them into the parent at the answer position, either as one mean vector over problems or as each problem’s own activations (“per-prompt states”); the same with 32 random nodes as control. “Format” = the fraction of outputs that are a number at all. Done for the elicit side (child into the elicit parent) and the teach side (taught child into its own parent: the blank base, or the format-installed base).

Elicit vs. teach. Figure 7, Table 10. **Elicit parent**: direct 0.000; self-chain **0.328** (= 0.484×0.668 , rewrite accuracy $\times$ symbol accuracy — this product held on four model variants); per-prompt states of the LoRA child 0.125, of the full-FT child 0.121 (random nodes: 0.000); the mean vector changes the output format only. **Teach parents**: blank base plus the same doses 0.000; blank plus chain 0.000 (a parser without an engine); taught donor into the blank base ≤ 0.004 in every condition, for all four taught children; the format-installed base with its own taught child’s per-prompt states 0.016. *Reading*: the elicit parent is one composition step or one activation patch away from the task because the machinery is there and only has

to be switched on; the teach parents have nothing underneath to switch on, so no patch from any taught model helps.

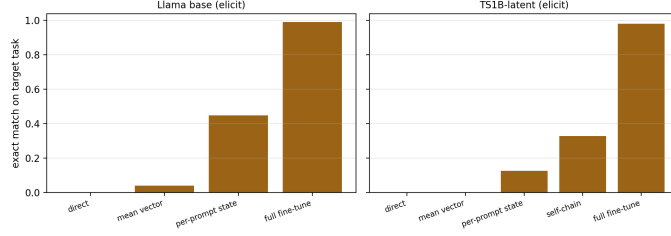


Figure 7: Intervention ladder on the elicit parent (weights frozen except the last bar).

Patient + intervention (32 circuit nodes)	condition	format	exact match
elicit parent + LoRA child’s mean vector	circuit nodes	0.164	0.000
elicit parent + LoRA child’s per-prompt states	circuit nodes	0.359	0.125
	random-32 nodes	0.051	0.000
elicit parent + full-FT child’s per-prompt states	circuit nodes	0.387	0.121
elicit parent self-chain (no donor)	own rewrite + “= ”	–	0.328
blank + binding dose, self-chain	parser without engine	–	0.000
blank + taught donor (LoRA 1M / LoRA 4M / full FT / converged)	every condition	≤ 0.02	≤ 0.004
format-installed base + its taught child’s per-prompt states	circuit nodes	0.836 (base 1.000)	0.016

Table 10: Zero-training interventions.

E. Every metric on the full-fine-tune endpoints

Both regimes under one method. The elicit twin and two teach twins, each fully fine-tuned with the identical recipe on the identical 4M file to convergence. The format-installed teach twin is the design-matched comparison (both parents already output numbers); the blank-base teach twin is the better-performing one. They agree on every row.

	elicit FT	teach FT, format-installed	teach FT, blank (conv.)	metric
exact match (0-shot)	0.986	0.424	0.819	M1
steps to stop (passes)	21,500 (0.7)	58,500 (1.9)	72,000 (2.3)	M1
held-out test loss (nats/token)	0.015	0.639	0.154	M1
circuit overlap with the elicit-FT circuit	–	0.306	0.333	M2
circuit overlap with the LoRA-elicited child	0.641	0.333	0.333	M2
top-8 nodes sufficient / necessary	0.923 / 0.943	0.983 / 0.988	0.950 / 0.965	M3
operand-reading heads vs the parent’s engine (first / second)	0.694 / 0.857	0.879 / 0.659	0.763 / 0.658	M4
gradient norm, first 1% $\rightarrow$ last 10% of steps	4.80 $\rightarrow$ 1.39 (decays 3.4 $\times$)	2.69 $\rightarrow$ 3.77 (grows 1.4 $\times$)	2.30 $\rightarrow$ 3.43 (grows 1.5 $\times$)	M6
cumulative gradient mass	44,021	378,170	353,277 (+ 30,954)	M6
relative weight write $\ \Delta W\ /\ W\ $	0.050	0.086	0.102	M7
PC1 of the state shift (after removing answer content)	0.056 (0.061)	0.567 (0.561)	0.526 (0.528)	M8
answer settled depth, median (quartiles)	layer 14 (14–14)	layer 15 (15–15)	layer 15 (15–15)	M10
J-space rank of the answer at layers 12 / 13 / 14	338 / 8 / 1	27,044 / 15,058 / 961	4,123 / 1,832 / 138	M10
child’s states patched into its own parent, per-prompt (mean)	0.121 (format only)	0.016 (0.000)	0.004 (0.000)	M11

Table 11: The full-FT endpoints, one row per metric.

Notes on measurement

- **Teach-side metrics were re-measured on performing taught models.** The first taught endpoint (LoRA 1M, exact match 0.093) had not learned the task. Every teach-side metric

was re-run on the LoRA-4M and the three full-FT taught endpoints, with the full-FT elicited endpoint as the method control. Three earlier readings changed and are stated in their revised form above: taught operand roles are mostly shared rather than absent (M4), the circuit difference is partial rather than total (M2), and the size of the taught state shift depends on the fine-tuning method (M8).

- **Answer depth is scored on the first digit token.** Scoring the raw first token credited a taught child with layer-5 formation on 48 problems — all subtractions with a negative answer, whose first token is the minus sign, produced by a comparison circuit. With the sign fed in first the effect vanishes; the other models were unaffected.
- **Construction controls.** The same doses on the blank twin change nothing on the target (0.000 in every probe); a sum-only binding dose leaves “difference” unbound; the composition law self-chain = rewrite $\times$ compute held on four variants.